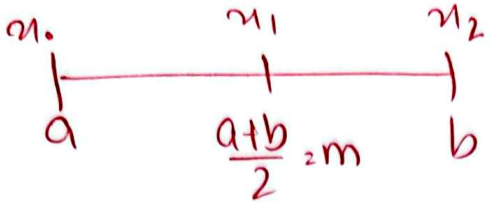
2b)




$$\sigma_1 = \int_a^b l_1(x)\, dx.$$

$$= \int_a^b \frac{x-x_0}{x_1-x_0} \times \frac{x-x_2}{x_1-x_2}\, dx.$$

$$= \int_a^b \frac{x-a}{m-a} \times \frac{x-b}{m-b}\, dx.$$

$$= \frac{1}{(m-a)(m-b)} \int_a^b (x-a)(x-b)\, dx$$

$$= \frac{1}{(m-a)(m-b)} \int_a^b x^2 - ax - bx + ab\, dx.$$

$$= \frac{1}{(m-a)(m-b)} \left[ \frac{x^3}{3} - \frac{ax^2}{2} - \frac{bx^2}{2} + abx \right]_a^b$$

$$= \frac{1}{(m-a)(m-b)} \left( \frac{b^3}{3} - \frac{ab^2}{2} - \frac{b^3}{2} + ab^2 - \frac{a^3}{3} + \frac{a^3}{2} + \frac{a^2b}{2} - a^2b \right)$$

$$= \frac{1}{\left(\frac{a+b}{2}-a\right)\left(\frac{a+b}{2}-b\right)} \left( \frac{2b^3 - 3ab^2 - 3b^3 + 6ab^2 - 2a^3 + 3a^3 + 3a^2b - 6a^2b}{6} \right)$$

$$= \frac{1}{\left(\frac{a+b-2a}{2}\right)\left(\frac{a+b-2b}{2}\right)} \left( \frac{-b^3 + 3ab^2 + a^3 - 3a^2b}{6} \right)$$

CamScanner

$$= \frac{1}{\left(\frac{b-a}{2}\right)\left(\frac{a-b}{2}\right)} \left\{ \frac{-(b^3 - 3ab^2 - a^3 + 3a^2b)}{6} \right\}$$

$$= \frac{4}{(b-a)(b-a)} \times \left\{ \frac{b^3 - a^3 - 3ab(b-a)}{6} \right\}$$

$$= \frac{4(b-a)^3}{6(b-a)^2}$$

$$= \frac{2}{3}(b-a)$$

CamScanner